

Planar Laser-Induced Fluorescence Imaging of OH Radicals (OH-PLIF) in Hypervelocity Combustion with an Intensified CCD System

The utility of future hypersonic air-breathing propulsion systems will depend on the efficient injection, mixing, and combustion processes in the high-speed combustion chamber. For scramjet engines, for example, these processes have to be optimized for flight speeds in excess of Mach 8. Most combustion research to date has focused on speeds much lower than this. Due to the large enthalpies associated with high-Mach flight, only impulse facilities are capable of providing the required total temperature and hypervelocity to adequately replicate the combustor environment. Expansion tubes and reflected shock tunnels are two such impulse environments where hypervelocity testing can be conducted. Expansion tubes have the advantage of less dissociated chemical species being formed, as well as more accurate simulation of combustion chemistry, including ignition delay and reaction times. However, the test times possible in expansion tubes are shorter than those of reflected shock tunnels.

Stanford Expansion Tube Facility

The hypervelocity combustion studies of Drs. Adela Ben-Yakar and Ronald K. Hanson (<http://navier.stanford.edu/hanson>) are conducted at the Stanford expansion tube facility of the High Temperature Gasdynamics Laboratory (HTGL). This facility is one of the few that can provide a wide range of total-enthalpy conditions, including 4 to 6 MJ/kg air to replicate Mach 10 to Mach 13 conditions. The expansion tube is 12 m in length with an inner diameter of 89 mm. It consists of three sections: driver, driven, and expansion. The driver section is filled with pressurized helium gas and is separated by a double diaphragm from the lower pressure-driven section, which is filled with the test gas. A run is initiated by bursting the diaphragms to form a shock wave that propagates into the test gas, producing intermediate velocity with increased pressure and temperatures. The shocked test gas is then accelerated by an area-expansion process into the expansion section, with both a higher stagnation enthalpy and a higher effective reservoir pressure than the shock tube flow from which it originated. A schematic of this setup is shown in Figure 1.

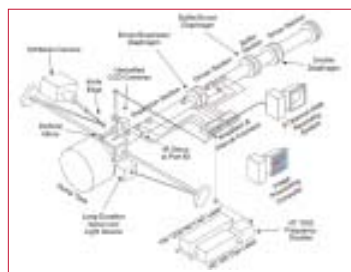


Figure 1: Schematic of the Stanford expansion tube facility used in these experiments.

The experiments performed by Ben-Yakar in the expansion tube used simultaneous Schlieren imaging and planar laser-induced fluorescence of OH radicals (OH-PLIF), formed by autoignition of a hydrogen jet, to obtain information on the location of the shock waves and the region of combustion during hypervelocity. Application of these two nonintrusive techniques permits unique and crucial visualization of the behavior of complex reacting supersonic fuel flowfields. The OH-fluorescence intensity can be associated with the OH mole fraction. At the combustion pressures obtained in this work, the fluorescence signal can be modeled briefly as

$$S_f \propto X_{OH} \left[\frac{f_{J''}}{T^{1/2}} \right]$$

where X_{OH} is the OH mole fraction and $f_{J''}$ is the Boltzmann fraction of OH molecules in the absorbing state.



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In this particular absorption transition — the Q1(7) transition of the $A^2\Sigma^+ \leftarrow X^2\Pi(1,0)$ band of OH, located at 283.31 nm — the effect of temperature plays only a small part in signal interpretation for regions containing OH. Fluorescence intensity, meanwhile, can be used as a direct indicator of OH mole fraction.

Experiment Setup

The Stanford HTGL investigators employ a Princeton Instruments PI-MAX intensified CCD (ICCD) camera system for the OH-PLIF experiment. This system is ideally suited for the application because of its high QE in the lower-wavelength region (200 to 500 nm) and fast gating. The system's nanosecond intensifier gating speeds and efficient on/off gating ratio of $10^7:1$ permit weak OH- fluorescence images to be distinguished from the strong background of the combustion flame.

The frequency-doubled output of a dye laser pumped by a pulsed Nd:YAG laser creates the required laser sheet. For OH-PLIF transitions near 283 nm, Rhodamine 590 dye is used with pulse energies of about 8 mJ. At the viewing section, the laser sheet is approximately 0.5-mm thick and 3-cm wide. The fluorescence signal is collected through the same exit window as that of the Schlieren system. A 5-cm diameter dichroic mirror is mounted at 45 degrees to the optical axis perpendicular to the exit window to separate the signals. The dichroic, which is designed for >99% reflectivity between 300 and 320 nm, is transparent to the Schlieren beam. The OH-fluorescence, however, is reflected and collected by the PI-MAX camera.

Experiment Results

Figure 2 shows simultaneous side-view OH-PLIF and Schlieren images of hydrogen injection into a supersonic crossflow at Mach 10 conditions. The OH-PLIF image displays the regions containing OH molecules. The evolution of the reaction zone is in good agreement with the jet position seen with Schlieren imaging. A significant and uniform level of OH in the recirculation area confined by the separation wave upstream of the injector is apparent. A thin filament along the outer edge of the plume attached to the recirculation-zone ignition region follows. Further downstream, a decrease in OH fluorescence is visible as the mixture expands around the jet flowfield. This decrease could be due to lower local-mixture temperature or poor mixing of air with the hydrogen of the jet.

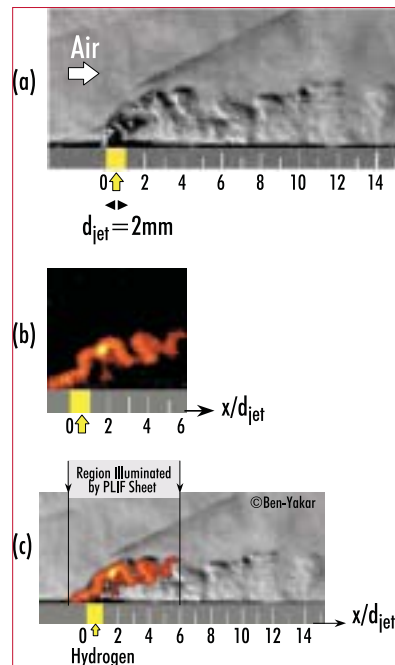


Figure 2: Simultaneous OH-PLIF and Schlieren images of hydrogen injected into a supersonic crossflow. Freestream conditions: air; burner entry Mach number = 3.46; temperature = 1300K; pressure = 0.32 atm; velocity = 2420 m/sec (Mach 10); jet-to-freestream flux ratio = 1.4.

- a) Schlieren image
- b) OH-PLIF image showing ignition and combustion regions of the jet in crossflow at high enthalpy
- c) OH-PLIF and Schlieren images overlaid



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Horizontally illuminated OH-PLIF images are then obtained (see Figure 3). These images show OH fluorescence around the jet, while the center of the plume has no OH formation. The results are consistent with those seen in Figure 2, showing that the OH concentration at the outer edge of the jet decreases as the jet moves downstream.

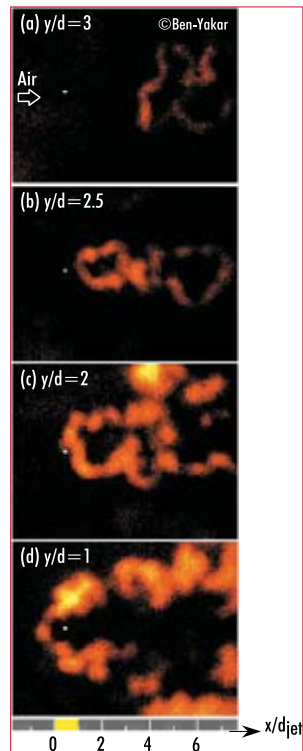


Figure 3: Top-view OH-PLIF images obtained at different heights (y/d = jet diameters) above the injection plate. Conditions are the same as in Figure 2. The white dots on the images indicate the center of the jet exit.

OH-PLIF imaging experiments at Mach 13 conditions show similar combustion characteristics as those at Mach 10. The pulsating nature of the upstream recirculation region can also be seen in Mach 13 conditions (see Figure 4).

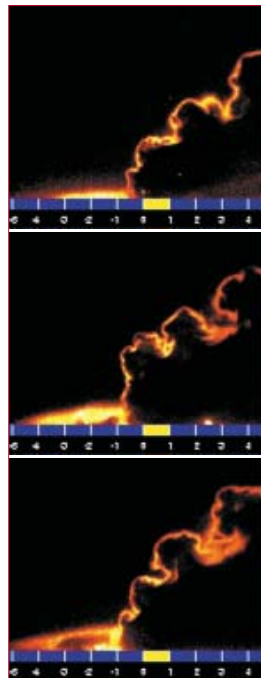


Figure 4: Three instantaneous sequential side-view OH-PLIF images at Mach 13 conditions. Conditions: pure oxygen; burner entry Mach number = 4.7; temperature = 1300K; pressure = 0.05 atm; velocity = 3300 m/sec (Mach 13).



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Conclusions

The Stanford HTGL scientists have been able to successfully generate high total-enthalpy flows simulating Mach 10 and Mach 13 conditions. The overlaid OH-PLIF and Schlieren images indicate the flame-holding capability of the hydrogen jet in air crossflow at different jet-to-freestream momentum flux ratios. The OH-PLIF images, obtained with an ICCD camera system, permit visualization of combustion that shows OH fluorescence at the recirculation region upstream from the jet and along the outer edge of the jet plume. These results and the techniques used to acquire them will contribute greatly to the successful development of future hypersonic airbreathing propulsion systems.

PI-MAX ICCD Cameras

Princeton Instruments' PI-MAX ICCD cameras offer better temporal resolution (gate widths <2 ns), superior background discrimination (MCP Bracket Pulsing™ with on/off ratios >107:1 in the UV), and greater sensitivity (QE >40%) than earlier generation ICCDs. PI-MAX systems employ an integrated Programmable Timing Generator™ that allows the user to set all gating parameters under GUI-software control. WinSpec and WinView software packages provide stand-alone 32-bit spectral and image acquisition, display, processing, and archiving functions. Each software package is customized to provide maximum ease and versatility for spectroscopic and imaging applications, respectively.

The optimized PI-MAX_{MG} ICCD system utilizes novel technology to achieve gating speeds of less than 9 ns, while preserving the highest possible QE of the photocathode, including crucial sensitivity in lower-wavelength regions of the spectrum.

References:

1. A. Ben-Yakar and R.K. Hanson. *Hypervelocity combustion studies using simultaneous OH-PLIF and Schlieren imaging in an expansion tube.* AIAA Paper 99-2453, 35th AIAA Joint Propulsion Conference, Los Angeles, CA, June 1999.
2. A. Ben-Yakar. *Experimental investigation of mixing and ignition of transverse jets in supersonic crossflows.* Ph.D. Thesis, Department of Mechanical Eng., Stanford University, December 2000.

PLIF images courtesy of Drs. Adela Ben-Yakar and Ronald K. Hanson (Stanford HTGL).



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